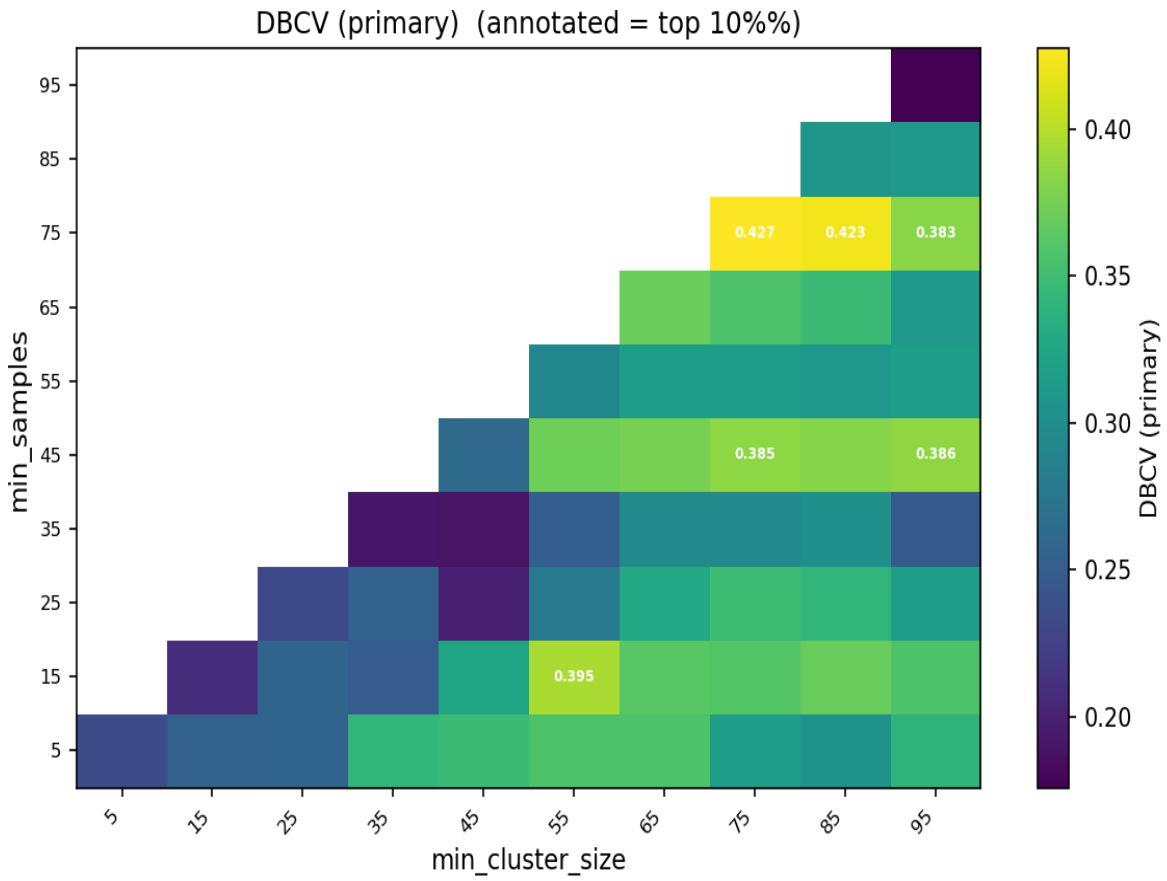
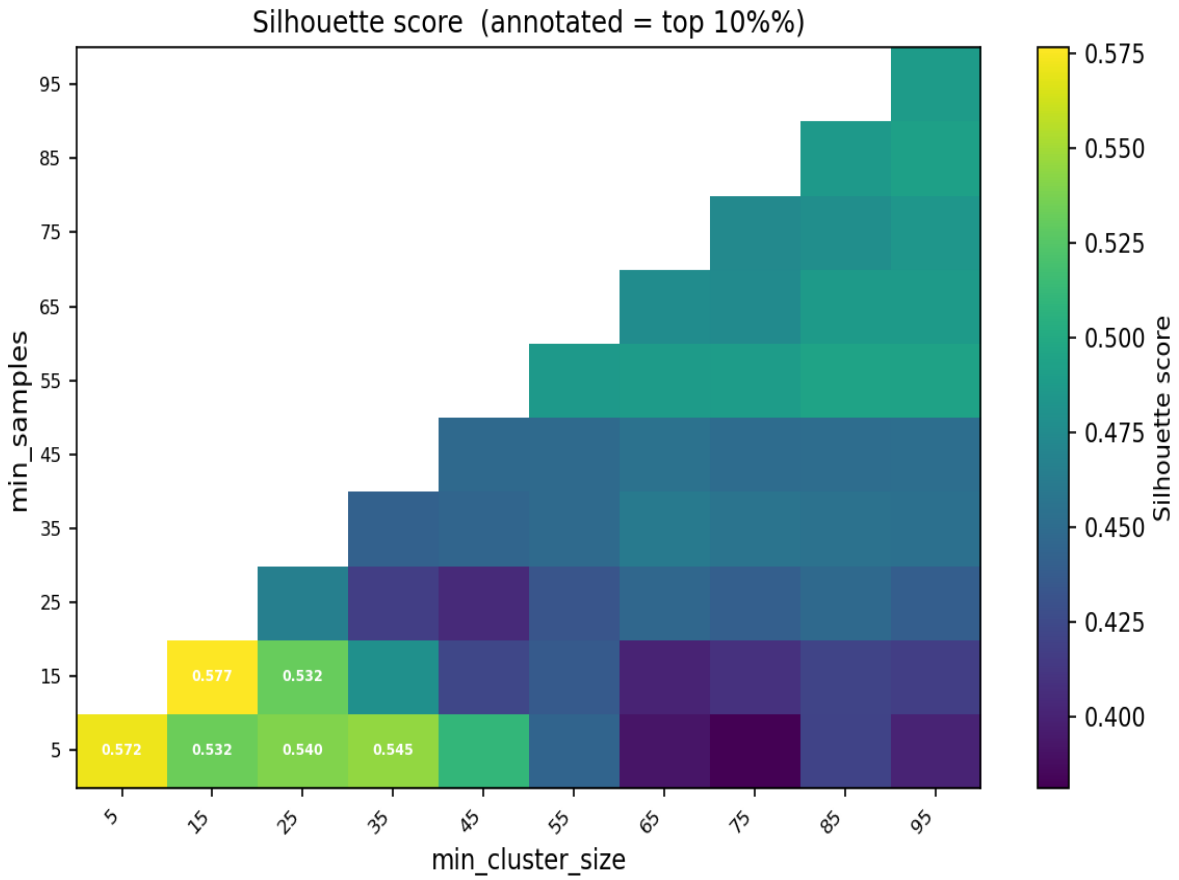
FragTree — Coarse Grid Search Report

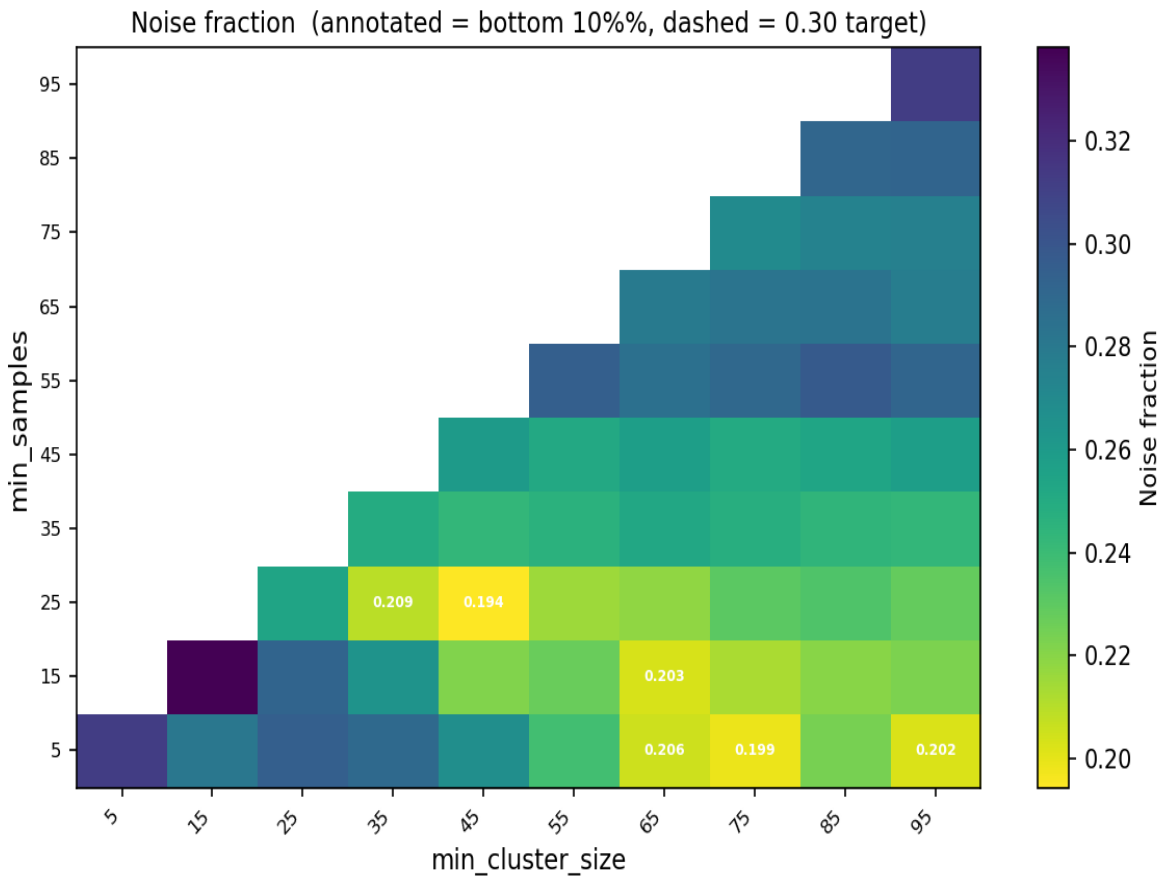
Generated: 2026-04-10 12:16:40 | n_molecules: 30,494

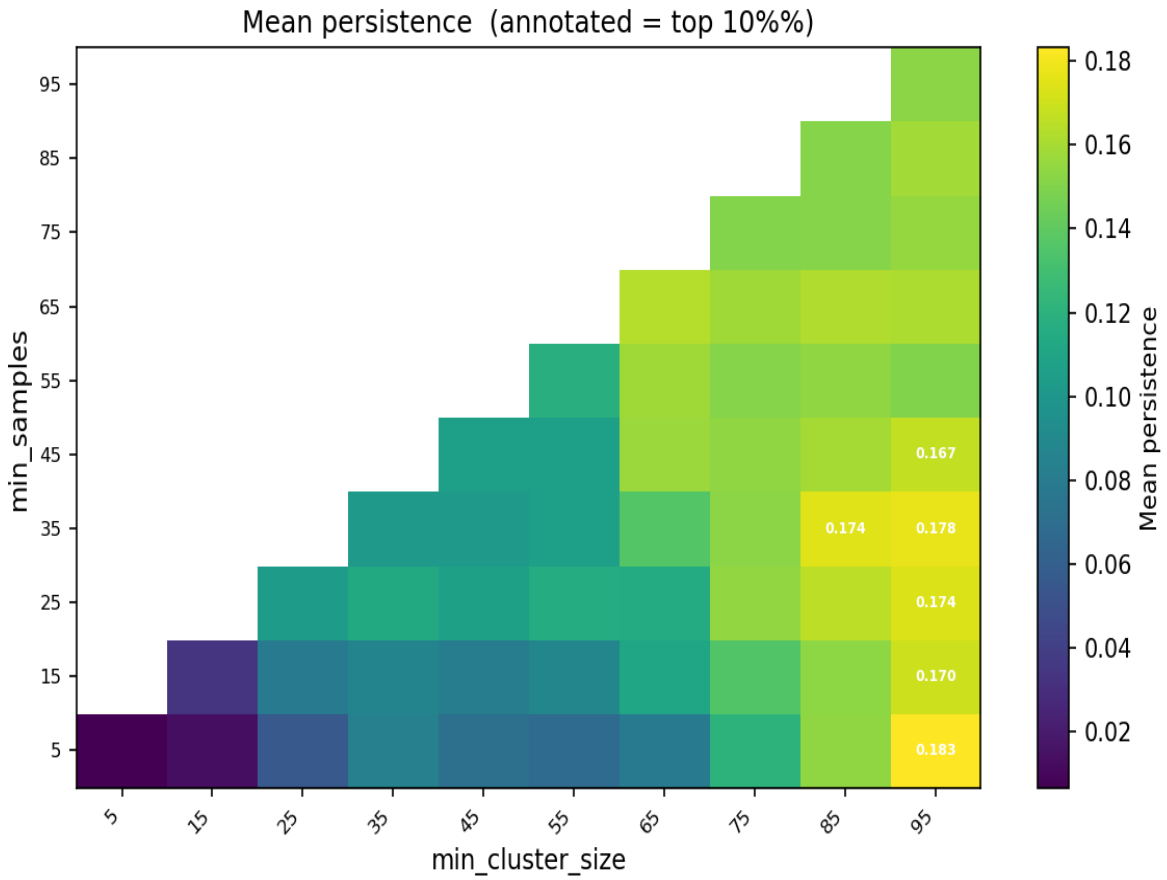
Parameter Sweep Heatmaps

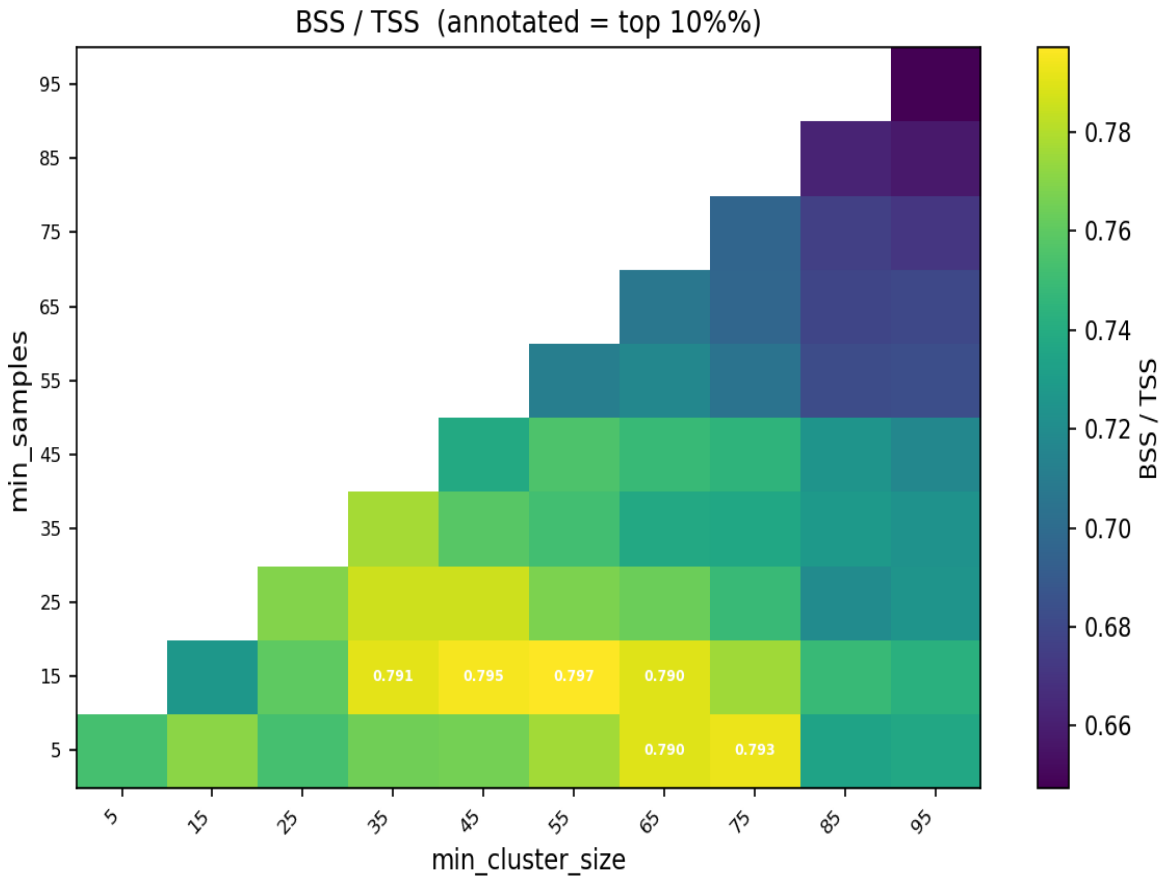
Each cell shows the metric value for a (min_cluster_size, min_samples) pair. Maximize DBCV, silhouette, mean persistence, and BSS/TSS. Minimize noise fraction (keep below 0.30). Cells with values in the top 10% are annotated with their score.




Silhouette score (annotated = top 10%%)
min_samples
min_cluster_size
Silhouette score
0.572
0.532
0.540
0.545
0.577
0.532
5
15
25
35
45
55
65
75
85
95
0.575
0.550
0.525
0.500
0.475
0.450
0.425
0.400


Noise fraction (annotated = bottom 10%%, dashed = 0.30 target)
min_samples
min_cluster_size
Noise fraction
0.209
0.194
0.203
0.206
0.199
0.202
5
15
25
35
45
55
65
75
85
95
0.20
0.22
0.24
0.26
0.28
0.30
0.32


Mean persistence (annotated = top 10%%)
min_samples
min_cluster_size
Mean persistence
95
85
75
65
55
45
35
25
15
5
0.167
0.174
0.178
0.174
0.170
0.183
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02


BSS / TSS (annotated = top 10%%)
min_samples
min_cluster_size
BSS / TSS
95
85
75
65
55
45
35
25
15
5
0.791
0.795
0.797
0.790
0.790
0.793
0.78
0.76
0.74
0.72
0.70
0.68
0.66

Top-10% Summary Table

A parameter pair is included if it falls in the top 10% of ANY metric (or bottom 10% for noise fraction, where lower is better). Sorted by DBCV descending.

cs	ms	DBCV	Silhouette	Noise frac	Mean persistence	BSS/TSS	N clusters
75	75	0.4275	0.4730	0.269	0.1502	0.6962	46
85	75	0.4225	0.4779	0.274	0.1513	0.6760	44
55	15	0.3953	0.4363	0.227	0.0880	0.7971	96
95	45	0.3859	0.4519	0.257	0.1668	0.7168	53
75	45	0.3847	0.4500	0.250	0.1542	0.7445	59
95	75	0.3832	0.4836	0.276	0.1555	0.6713	41
65	15	0.3625	0.4009	0.203	0.1110	0.7900	74
65	5	0.3575	0.3920	0.206	0.0785	0.7897	93
95	15	0.3571	0.4176	0.223	0.1698	0.7424	60
35	5	0.3421	0.5451	0.289	0.0834	0.7645	216
95	5	0.3400	0.4005	0.202	0.1833	0.7372	53
45	15	0.3239	0.4230	0.222	0.0807	0.7950	116
75	5	0.3173	0.3812	0.199	0.1211	0.7927	74
95	25	0.3169	0.4400	0.228	0.1741	0.7249	57
85	35	0.3022	0.4546	0.244	0.1744	0.7279	56
25	5	0.2567	0.5399	0.295	0.0559	0.7527	290
25	15	0.2566	0.5315	0.292	0.0795	0.7599	214
15	5	0.2553	0.5325	0.281	0.0136	0.7713	460
35	25	0.2545	0.4175	0.209	0.1139	0.7859	110
35	15	0.2490	0.4795	0.264	0.0862	0.7909	168
95	35	0.2473	0.4535	0.243	0.1776	0.7237	54
5	5	0.2343	0.5716	0.312	0.0067	0.7531	1008
15	15	0.2089	0.5767	0.338	0.0340	0.7266	326
45	25	0.1979	0.4047	0.194	0.1068	0.7855	89